

```
In [5]: import torch
import numpy as np
from PIL import Image
# from basicsr.archs.rrdbnet_arch import RRDBNet
# from realesrgan import RealESRGANer
import matplotlib.pyplot as plt
import cv2
```

```
In [ ]: models_path = "RealESRGAN_x4plus.pth"
state_dict = torch.load(models_path, map_location="cpu")["params_ema"]

chosed_model= RRDBNet(num_in_ch=3, num_out_ch=3, num_feat=64, num_block=23, num_grow_ch=32, scale=4 )
chosed_model.load_state_dict(state_dict, strict=True)

upsampler = RealESRGANer(scale=4, model_path=models_path, model=chosed_model,
                          tile=0, pre_pad=0, half=False)

image = Image.open("bald.jpg").convert("RGB")
img = image.resize((100, 72))
# img = Image.open("Training_7602986.jpg").convert("RGB")
img = np.array(img)

output, _ = upsampler.enhance(img, outscale=4)

output_img = Image.fromarray(output)
output_img.save("output2.jpg")

bild = output_img.resize((64,64))

plt.figure(figsize=(6, 10))

plt.subplot(3, 1, 1)
plt.imshow(img)

plt.subplot(3, 1, 2)
plt.imshow(output_img)

plt.subplot(3, 1, 3)
plt.imshow(image)
```

```
plt.tight_layout()  
plt.show()
```





Demo of GFPGAN with some Examples

```
In [14]: image_100 = Image.open("test_0316_aligned.jpg").convert("RGB")
output_image = Image.open("test_0316_alignedV2.jpg").convert("RGB")
#4x scaled
output_image2 = Image.open("test_0316_alignedV22.jpg").convert("RGB")
# img = Image.open("Training_7602986.jpg").convert("RGB")

resized_output = output_image.resize((64,64))
resized_img = image_100.resize((64,64))
resized_output2 = output_image2.resize((64,64))

#####
image_200 = Image.open("test_0316_aligned.jpg").convert("RGB")
output_imageGAN = Image.open("test_0316_alignedGAN.jpg").convert("RGB")
#4x scaled
output_imageGAN2 = Image.open("test_0316_alignedGAN2.jpg").convert("RGB")

resized_outputGAN = output_imageGAN.resize((64,64))
resized_imgGAN = image_200.resize((64,64))
resized_outputGAN2 = output_imageGAN2.resize((64,64))

plt.figure(figsize=(12, 20))
```

```
plt.subplot(1, 3, 1)
plt.imshow(resized_img)
plt.title("Original")

plt.subplot(1, 3, 2)
plt.imshow(resized_output)
plt.title("GAN")

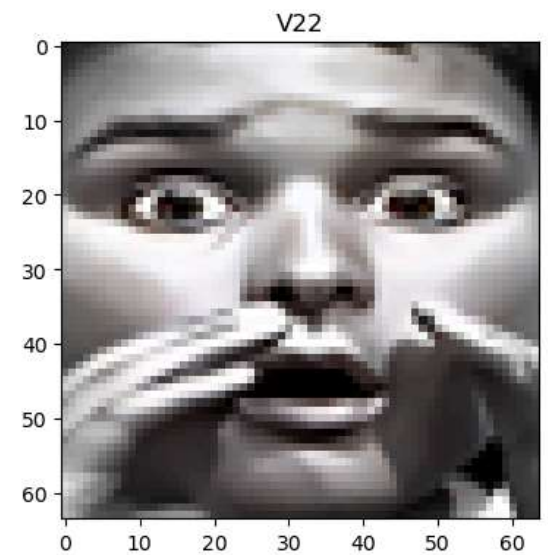
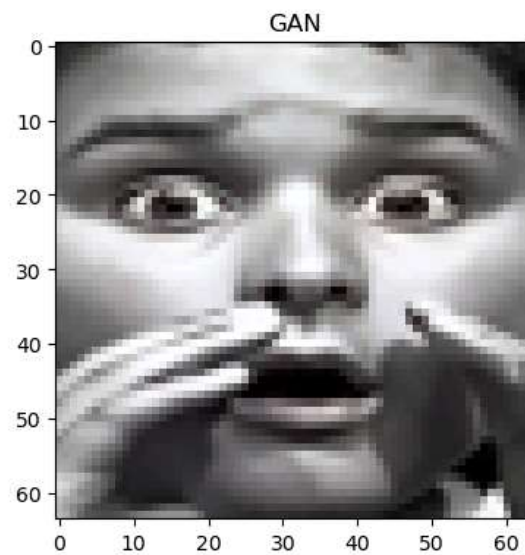
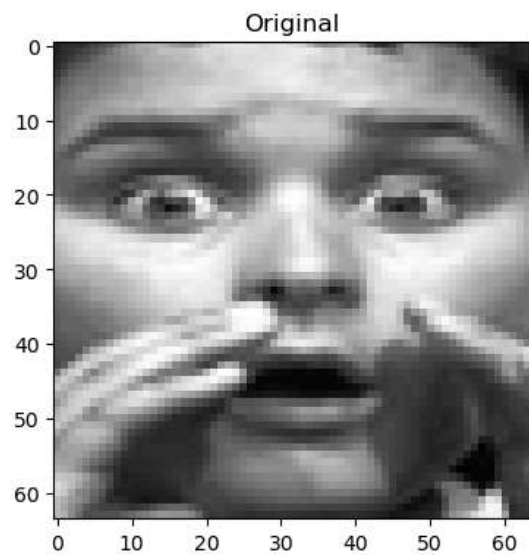
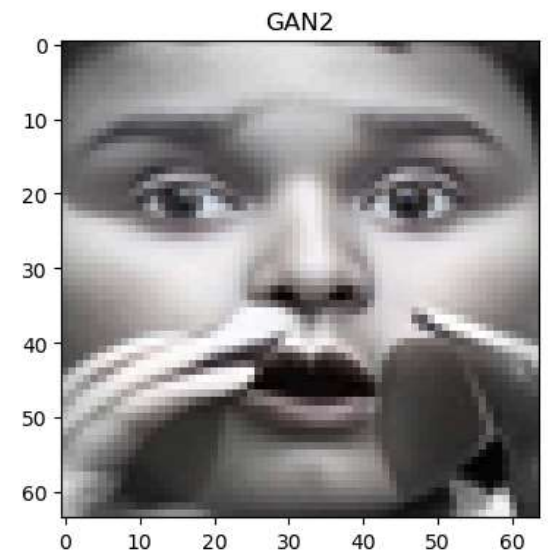
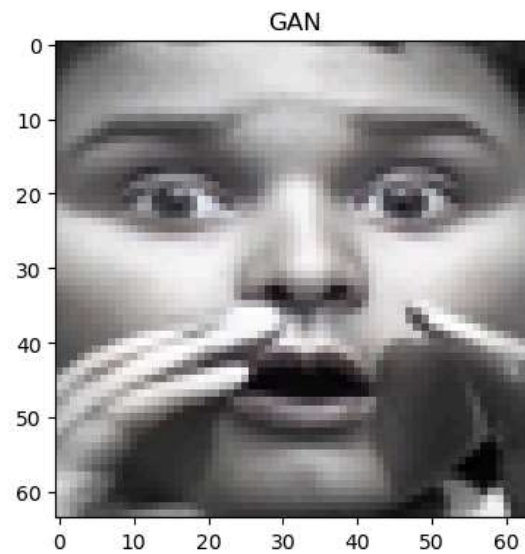
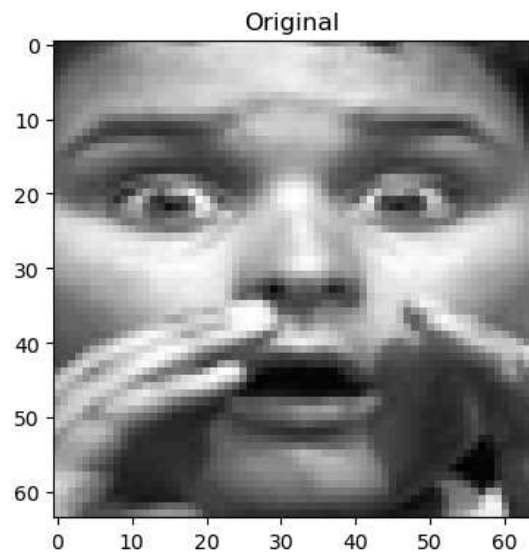
plt.subplot(1, 3, 3)
plt.imshow(resized_output2)
plt.title("V22")

plt.subplot(2, 3, 1)
plt.imshow(resized_imgGAN)
plt.title("Original")

plt.subplot(2, 3, 2)
plt.imshow(resized_outputGAN)
plt.title("GAN")

plt.subplot(2, 3, 3)
plt.imshow(resized_outputGAN2)
plt.title("GAN2")

plt.tight_layout()
plt.show()
```



```
In [17]: image_100 = Image.open("test_1013_aligned.jpg").convert("RGB")
         output_image = Image.open("test_1013_alignedV2.jpg").convert("RGB")
         # with the 4times scaled
```

```

output_image2 = Image.open("test_1013_alignedV22.jpg").convert("RGB")

# img = Image.open("Training_7602986.jpg").convert("RGB")

resized_output = output_image.resize((64,64))
resized_img = image_100.resize((64,64))
#with the 4times scaled
resized_output2 = output_image2.resize((64,64))
#####

image_200 = Image.open("test_1013_aligned.jpg").convert("RGB")
output_imageGAN = Image.open("test_1013_alignedGAN.jpg").convert("RGB")
#4x scaled
output_imageGAN2 = Image.open("test_1013_alignedGAN2.jpg").convert("RGB")

resized_outputGAN = output_imageGAN.resize((64,64))
resized_imgGAN = image_200.resize((64,64))
resized_outputGAN2 = output_imageGAN2.resize((64,64))

plt.figure(figsize=(12, 20))

plt.subplot(1, 3, 1)
plt.imshow(resized_img)
plt.title("Original")

plt.subplot(1, 3, 2)
plt.imshow(resized_output)
plt.title("GAN")

plt.subplot(1, 3, 3)
plt.imshow(resized_output2)
plt.title("V22")

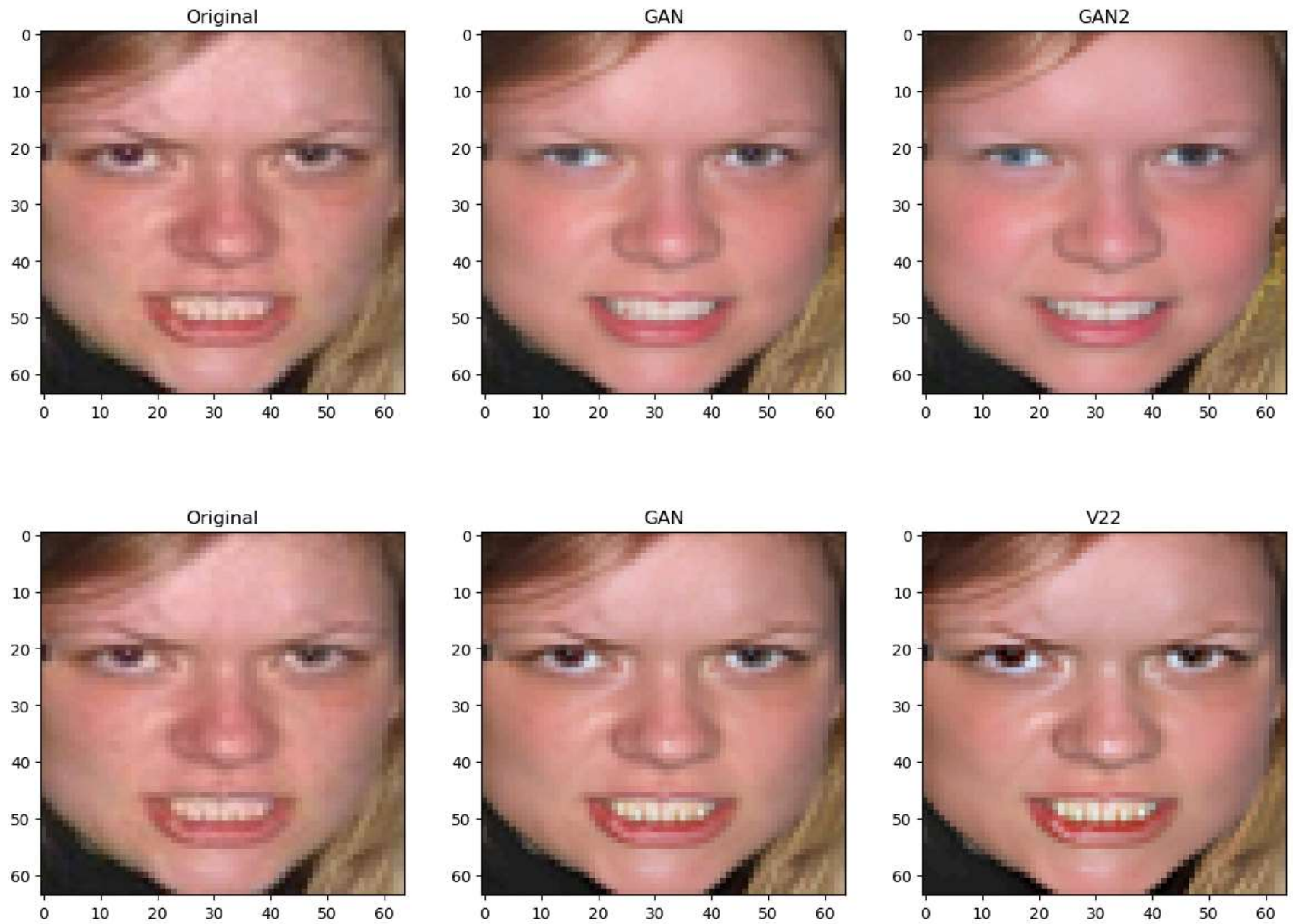
plt.subplot(2, 3, 1)
plt.imshow(resized_imgGAN)
plt.title("Original")

plt.subplot(2, 3, 2)
plt.imshow(resized_outputGAN)
plt.title("GAN")

```

```
plt.subplot(2, 3, 3)
plt.imshow(resized_outputGAN2)
plt.title("GAN2")
```

```
plt.tight_layout()
plt.show()
```

```
In [21]: image_100 = Image.open("test_0149_aligned.jpg").convert("RGB")
         output_image = Image.open("test_0149_alignedV2.jpg").convert("RGB")
         #
```

```

output_image2 = Image.open("test_0149_alignedV22.jpg").convert("RGB")
# img = Image.open("Training_7602986.jpg").convert("RGB")

resized_output = output_image.resize((64,64))
resized_img = image_100.resize((64,64))

resized_output2 = output_image2.resize((64,64))
#####

image_200 = Image.open("test_0149_aligned.jpg").convert("RGB")
output_imageGAN = Image.open("test_0149_alignedGAN.jpg").convert("RGB")
#4x scaled
output_imageGAN2 = Image.open("test_0149_alignedGAN2.jpg").convert("RGB")

resized_outputGAN = output_imageGAN.resize((64,64))
resized_imgGAN = image_200.resize((64,64))
resized_outputGAN2 = output_imageGAN2.resize((64,64))


plt.figure(figsize=(12, 20))

plt.subplot(1, 3, 1)
plt.imshow(resized_img)
plt.title("Original")

plt.subplot(1, 3, 2)
plt.imshow(resized_output)
plt.title("GAN")

plt.subplot(1, 3, 3)
plt.imshow(resized_output2)
plt.title("GAN2")


plt.subplot(2, 3, 1)
plt.imshow(resized_imgGAN)
plt.title("Original")

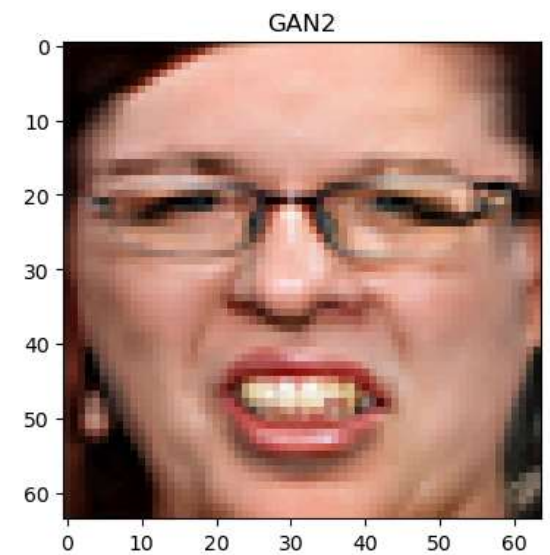
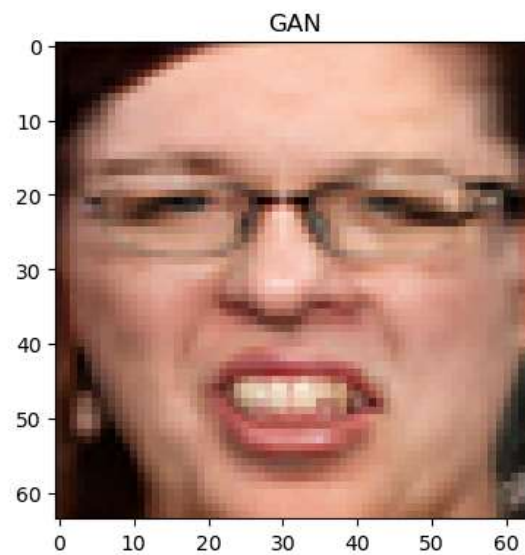
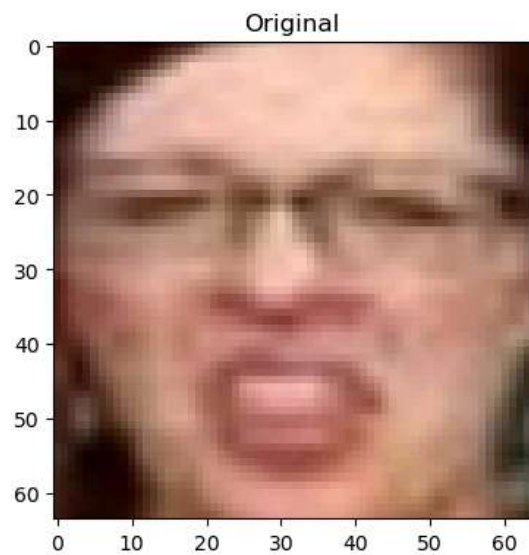
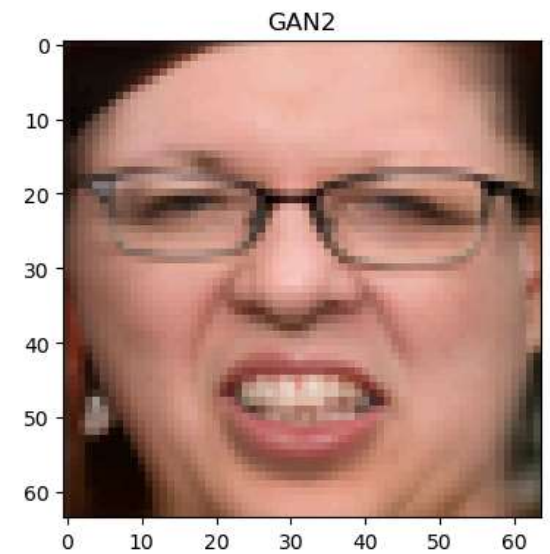
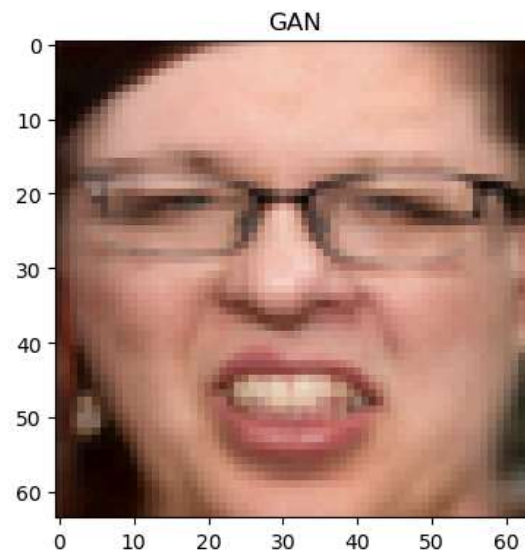
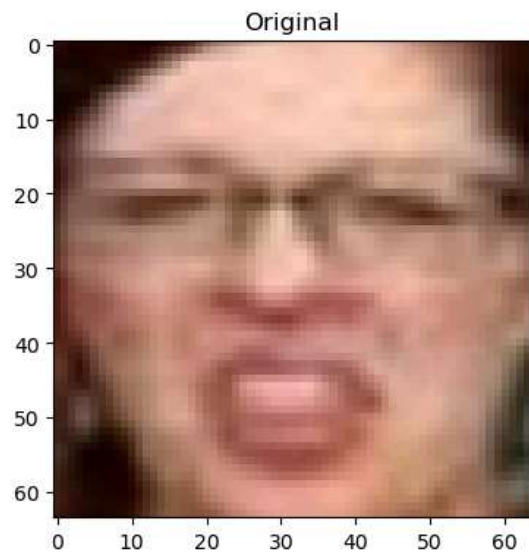
plt.subplot(2, 3, 2)
plt.imshow(resized_outputGAN)

```

```
plt.title("GAN")

plt.subplot(2, 3, 3)
plt.imshow(resized_outputGAN2)
plt.title("GAN2")

plt.tight_layout()
plt.show()
```



```
In [20]: image_100 = Image.open("test_0012_aligned.jpg").convert("RGB")
         output_image = Image.open("test_0012_alignedV2.jpg").convert("RGB")
         #
```

```

output_image2 = Image.open("test_0012_alignedV22.jpg").convert("RGB")

# img = Image.open("Training_7602986.jpg").convert("RGB")

resized_output = output_image.resize((64,64))
resized_img = image_100.resize((64,64))

resized_output2 = output_image2.resize((64,64))

#####

image_200 = Image.open("test_0012_aligned.jpg").convert("RGB")
output_imageGAN = Image.open("test_0012_alignedGAN.jpg").convert("RGB")
#4x scaled
output_imageGAN2 = Image.open("test_0012_alignedGAN2.jpg").convert("RGB")

resized_outputGAN = output_imageGAN.resize((64,64))
resized_imgGAN = image_200.resize((64,64))
resized_outputGAN2 = output_imageGAN2.resize((64,64))

plt.figure(figsize=(12, 20))

plt.subplot(1, 3, 1)
plt.imshow(resized_img)
plt.title("Original")

plt.subplot(1, 3, 2)
plt.imshow(resized_output)
plt.title("GAN")

plt.subplot(1, 3, 3)
plt.imshow(resized_output2)
plt.title("V22")

plt.subplot(1, 3, 3)
plt.imshow(resized_output2)
plt.title("GAN2")

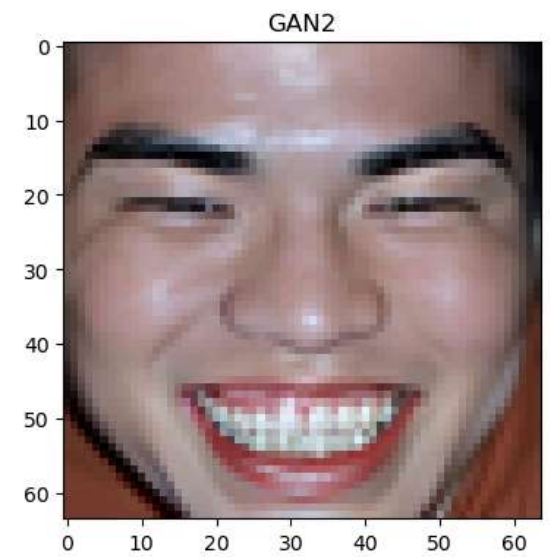
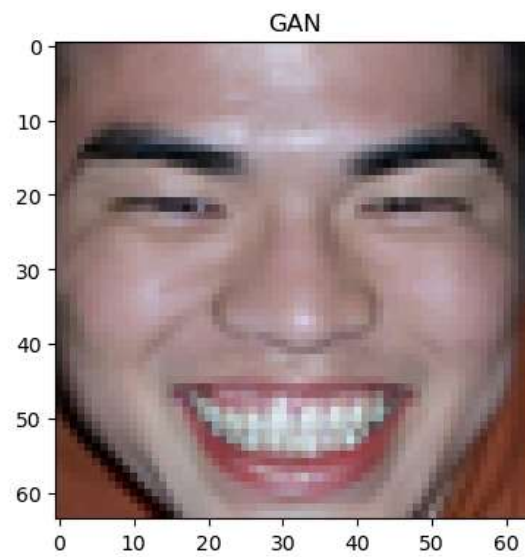
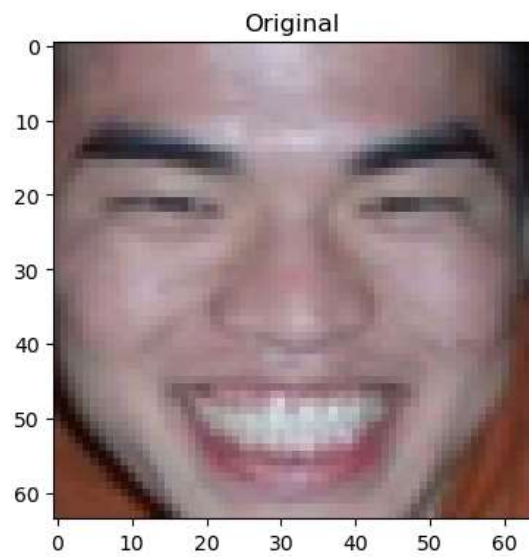
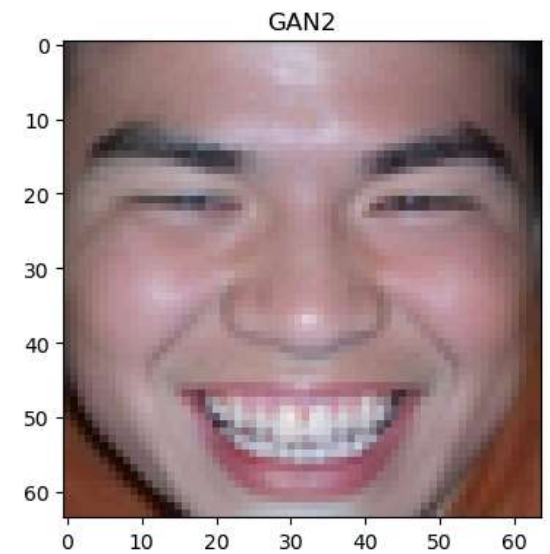
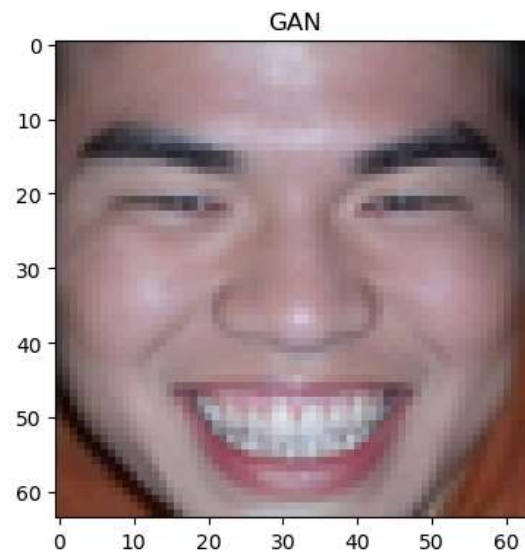
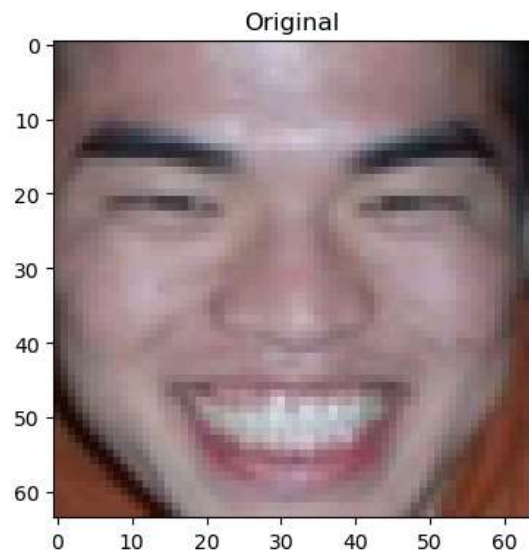
```

```
plt.subplot(2, 3, 1)
plt.imshow(resized_imgGAN)
plt.title("Original")

plt.subplot(2, 3, 2)
plt.imshow(resized_outputGAN)
plt.title("GAN")

plt.subplot(2, 3, 3)
plt.imshow(resized_outputGAN2)
plt.title("GAN2")

plt.tight_layout()
plt.show()
```

```
In [19]: image_100 = Image.open("Training_7602986.jpg").convert("RGB")
         output_image = Image.open("Training_7602986V2.jpg").convert("RGB")
         #
```

```

output_image2 = Image.open("Training_7602986V22.jpg").convert("RGB")
# img = Image.open("Training_7602986.jpg").convert("RGB")

resized_output = output_image.resize((64,64))
resized_img = image_100.resize((64,64))
resized_output2 = output_image2.resize((64,64))

#####

image_200 = Image.open("Training_7602986.jpg").convert("RGB")
output_imageGAN = Image.open("Training_7602986GAN.jpg").convert("RGB")
#4x scaled
output_imageGAN2 = Image.open("Training_7602986GAN2.jpg").convert("RGB")

resized_outputGAN = output_imageGAN.resize((64,64))
resized_imgGAN = image_200.resize((64,64))
resized_outputGAN2 = output_imageGAN2.resize((64,64))

plt.figure(figsize=(12, 20))

plt.subplot(1, 3, 1)
plt.imshow(resized_img)
plt.title("Original")

plt.subplot(1, 3, 2)
plt.imshow(resized_output)
plt.title("GAN")

plt.subplot(1, 3, 3)
plt.imshow(resized_output2)
plt.title("GAN2")

plt.subplot(2, 3, 1)
plt.imshow(resized_imgGAN)
plt.title("Original")

```



```
plt.subplot(2, 3, 2)
plt.imshow(resized_outputGAN)
plt.title("GAN")

plt.subplot(2, 3, 3)
plt.imshow(resized_outputGAN2)
plt.title("GAN2")

plt.tight_layout()
plt.show()
```



```
In [18]: image_100 = Image.open("train_00271_aligned.jpg").convert("RGB")
         output_image = Image.open("train_00271_alignedGAN.jpg").convert("RGB")
         output_image2 = Image.open("train_00271_alignedGAN2.jpg").convert("RGB")
```

```

# img = Image.open("Training_7602986.jpg").convert("RGB")

resized_output = output_image.resize((64,64))
resized_img = image_100.resize((64,64))

resized_output2 = output_image2.resize((64,64))

#####

image_200 = Image.open("train_00271_aligned.jpg").convert("RGB")
output_imageGAN = Image.open("train_00271_alignedGAN.jpg").convert("RGB")
#4x scaled
output_imageGAN2 = Image.open("train_00271_alignedGAN2.jpg").convert("RGB")

resized_outputGAN = output_imageGAN.resize((64,64))
resized_imgGAN = image_200.resize((64,64))
resized_outputGAN2 = output_imageGAN2.resize((64,64))

plt.figure(figsize=(6, 10))

plt.subplot(1, 3, 1)
plt.imshow(resized_img)
plt.title("Original")

plt.subplot(1, 3, 2)
plt.imshow(resized_output)
plt.title("GAN")

plt.subplot(1, 3, 3)
plt.imshow(resized_output2)
plt.title("GAN2")

plt.subplot(2, 3, 1)
plt.imshow(resized_imgGAN)
plt.title("Original")

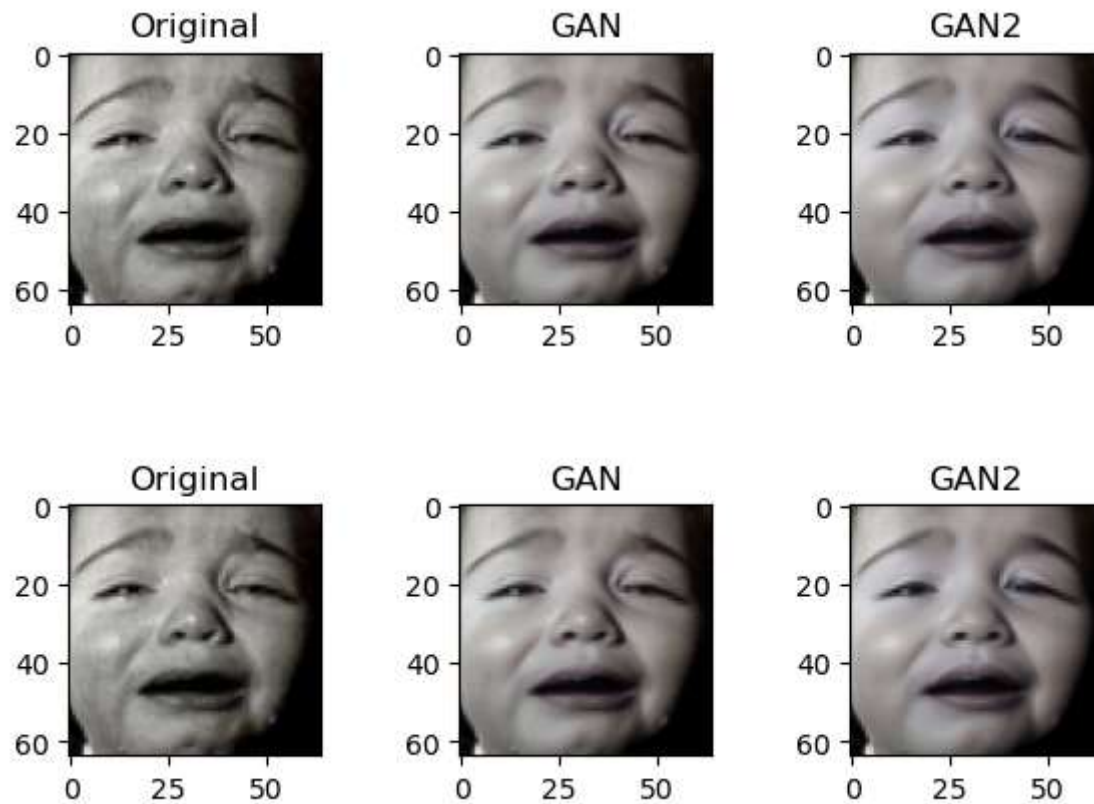
plt.subplot(2, 3, 2)
plt.imshow(resized_outputGAN)

```

```
plt.title("GAN")

plt.subplot(2, 3, 3)
plt.imshow(resized_outputGAN2)
plt.title("GAN2")

plt.tight_layout()
plt.show()
```



```
In [ ]: image_100 = Image.open("train_00271_aligned.jpg").convert("RGB")
output_image = Image.open("train_00271_alignedV2.jpg").convert("RGB")
output_image2 = Image.open("train_00271_alignedV22.jpg").convert("RGB")

# img = Image.open("Training_7602986.jpg").convert("RGB")

resized_output = output_image.resize((64,64))
resized_img = image_100.resize((64,64))
```

```
resized_output = output_image2.resize((64,64))

plt.figure(figsize=(6, 10))

plt.subplot(1, 3, 1)
plt.imshow(resized_img)
plt.title("Original")

plt.subplot(1, 3, 2)
plt.imshow(resized_output)
plt.title("GAN")

plt.subplot(1, 3, 3)
plt.imshow(resized_output2)
plt.title("GAN2")

plt.tight_layout()
plt.show()
```